Learning & Knowledge Suite: Build Specification (v3)

For the AI engineer building the product. Read this sheet first, then follow the order below.

1. What you are building

A learning platform for companies in any service industry. Every employee is mapped to a role; every role has required skills; every skill has a test. Test results give each person a level (0 to 5) and a rank for each skill. From the gaps the system builds a learning pathway, tracks participation and reminds until it is finished. Pathways link to internal career moves (up and sideways) and to certificates. Courses are generated with AI, some with AR or interactive simulation. Gamification and live leaderboards keep people engaged; reports can be exported; every workflow can be manual, automatic or hybrid; mandatory training such as POSH and DPDP can be proctored. A completely separate module, the Knowledge Hub, lets any team store, find, review and control SOPs, policies and company data, with versions, reminders, owner-controlled access and leader dashboards.

2. How to use this workbook

Sheet	What is in it	How you use it
01 Core Flow	The whole loop in 14 steps: who does what, what is automatic, what can be done by hand	Start here
02 Requirements	Every feature as a requirement with a test you can run (acceptance criteria), priority and screen	Your to-do list
03 Workflows	25 workflows, step by step, with default mode (manual, automated or hybrid)	Build the workflow engine from this
04 Skills Rules	Score to level table, ranking, gap, availability, pathway and career rules (the maths)	Rules the code must follow
05 to 08 (library)	247 skills, 187 roles for 15 industry groups, 1,871 role-skill requirements, 185 career moves	Seed data to load into the database
09 Assessment & Proctoring	Skill tests, proctoring levels, mandatory training, certificates	Build the test and certificate engines
10 AI Courses & AR	AI course pipeline, video, AR and interactive simulation, scenario file format	Build the AI studio
11 Gamification & Leaderboards	Points, levels, badges, challenges, real-time leaderboards	Build engagement features
12 Invites & Reminders	Invitations for scheduled sessions; completion reminders at a convenient time	Build notifications
13 Reports	32 standard reports with fields, filters and exports	Build the reports centre
14 Knowledge Hub	Design of the separate module: Library view, versions, reminders, access, dashboards	Build second, in parallel
14a to 14d (Hub)	44 requirements with tests; types and categories; 164 demo items; demo leader and function dashboard with live formulas	Requirements, lookup tables and expected numbers
15 to 19 (demo)	A working demo company: 36 employees, tests, levels, ranks, gaps, pathways, careers	Test data and a model that shows the expected numbers
20 Data Model	The records and fields	Create the database

21 AI & Security	AI guardrails, privacy, security and legal rules	Non-negotiable rules
22 UI Screens	Every screen with its purpose and the UI deck slide	Build the front end
23 Build & Tests	Build order, suggested stack and 20 end-to-end tests	Plan and verify
24 References	External facts used, with links	Check when unsure

3. Golden rules (apply to every screen and every feature)

Rule	What it means	Type
1. One main action per screen	Each screen has one big primary button. Everything else is secondary.	UI rule
2. Plain words	Say "Start learning", not "Initiate module". No jargon. Short sentences. Hindi and English.	UI rule
3. Always show what is next	Every home screen shows the next best step. No dead ends.	UI rule
4. Safe by default	Confirm before anything is sent to many people; show a preview; allow undo for 10 minutes.	UX safety
5. Humans decide	AI drafts and suggests. People approve content, scores in doubt, proctoring flags and certificates.	AI rule
6. Everything can be done by hand	Any automatic step has a manual way, and every workflow can be set to Manual, Automated or Hybrid.	Workflow rule
7. Everything is logged	Overrides, exports, deletions, role changes and approvals are written to the audit log.	Security rule
8. Settings, not hard-coded numbers	Every default in this workbook (thresholds, hours, caps, retention) lives in a settings table.	Engineering rule
9. Mobile first, accessible	Design for a low-cost phone first. Meet WCAG 2.1 AA.	UI rule
10. Privacy first	Collect only what is needed; show clear notices; follow the data-privacy rules in sheet 21.	Legal rule

4. Words used in this workbook

Word	Meaning	
Role	A job. Every employee has one primary role.	
Skill	One thing a person must be able to do. A role needs 10 to 16 skills.	
Level	0 to 5: none, awareness, working, proficient, advanced, expert. Level 3 is role-ready for most roles.	
Required level	The level a role needs for a skill.	
Gap	Required level minus current level, never below zero.	
Stack rank	Position of a person among all tested people for one skill (rank, out of how many, percentile).	

Pathway	A person's learning plan: waves of 3 skills; each skill has four steps: learn, practise, connect, prove.	
Career map	Next roles a person could move to (vertical = promotion, lateral = sideways) with readiness.	
Certification course	A course that ends in a pass mark and a certificate that can be verified.	
Workflow mode	Manual (a person does it), Automated (the system does it), Hybrid (the system does it after a person approves).	
MVP, V2, V3	Priority. MVP first, then V2, then V3.	

Everything in the demo sheets (15 to 19) is invented test data. Real people and companies are never used for development or testing.